

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC402 - ENVIRONMENTAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The unit of turbidity of water is _____.
a)JTU
b)NTU
c)PPM
d)All of above
- 2 Dosage of alum is decided by _____ test.
a)Hardness
b)Chloride
c)Jar
d)Char
- 3 Cleaning interval for rapid sand filter bed is _____.
a)3 – 4 months
b)6 – 8 months
c)1 – 2 year
d)24 hours
- 4 If the sedimentation tank is rectangular in shape having length L, width W and depth D, then for discharge (Q), the settling velocity of particle would be _____.
a) $Q / (W \times D)$
b) $Q / (L \times W)$
c) $Q / (D \times L)$
d) $Q / (L \times W \times D)$
- 5 BOD₅ is determined in laboratory at ____ degree celcius.
a) 15 b) 20 c) 25 d) 10
- 6 Screen chamber is provided to remove _____ from raw wastewater
a) Dissolved Solids
b) Volatile Solids
c) fixed solids
d) Suspended Solids

- 7 Ozone is present in _____ layer of Atmosphere
 a) troposphere
 b) mesosphere
 c) stratosphere
 d) thermosphere
- 8 Pick out the odd one with respect to classification of solid wastepaper. card board, wood, metal
 a) paper
 b) card board
 c) wood
 d) metal
- 9 Function of cascade aerator is to _____.
 a) remove iron and manganese
 b) to remove odors
 c) to increase oxygen content in water
 d) to mix chemicals if added in water e) all of above
- 10 _____ condition is most unfavourable for dispersion of air pollutants in the atmosphere. a) Inversion b) unstable c) Neutral d) all of above
- 11 The ultimate (Last) disposal option for solid waste management is _____. a) segregation b) Composting c) Land filling d) none of above
- 12 As per IS 10500 MPN should be _____.
 a) zero
 b) 10
 c) 20
 d) 100
- Q2. Solve the following.
- A) List population forecasting methods. Explain arithmetic mean method. The population of a town was 25,000 in 1991, 36,000 in 2001, and 47,000 in 2011. Estimate the population in 2031 using the **Arithmetic Increase Method**. 6
- B) Explain various types of water demands? Write fire demand formulae 6
- Q3. Solve the following.
- A) Design a clariflocculator to treat a flow of 10 MLD. Assume necessary data 6
- B) What is meant by back-washing? Explain how cleaning is performed in Rapid sand Filter with neat sketch 6
- Q4. Solve Any Two of the following.
- A) List out four common types of **water distribution system layouts**. Discuss any one in detail with neat diagram. 6
- B) Compare **gravity**, **pumping**, and **combined (gravity + pumping)** methods of water supply. Which is most preferred and why? 6

- C) Three pipes are connected **in series** to carry water between two reservoirs. Their **lengths and diameters** are as follows: 6

Pipe	Length (L) in meters	Diameter (D) in mm
A	400	300
B	600	250
C	500	200

Assume that the **friction factor (f)** is the same for all pipes, and that the **Darcy-Weisbach equation** applies.

Calculate the **equivalent length of a single pipe** of diameter **200 mm** that will produce the same head loss as the three pipes in series.

- Q5. Solve Any Two of the following.

- A) Draw a wastewater treatment flowsheet and tabulate functions of each unit. 6
- B) Define 'Solid Waste'. Classify the following solid wastes as **biodegradable or non-biodegradable**. Also suggest disposal methods for each category. 6
Items: 1) Food waste, 2) Plastic bottles, 3) Vegetable peels, 4) Broken glass, 5) Paper
- C) Define **BOD**. A wastewater sample was tested for BOD. The 5-day BOD at 20°C was found to be **120 mg/L**. If the reaction rate constant K_D at 20°C is **0.1 day⁻¹**, calculate the **ultimate BOD (BOD_u)** using the first-order BOD equation. 6

- Q6. Solve Any Two of the following.

- A) Define **air pollution** and classify the **types of air pollutants** with suitable examples. 6
- B) Explain **atmospheric stability** and its effect on **plume behavior**. Sketch and label any **five common plume types**. 6
- C) List **air pollution control devices** used for industrial air emissions and state working principle, advantages and disadvantages of gravity Gravity settling chamber with neat sketch. 6

*** End ***